Loïc Desgeorges

Maître de conférences (Associate Professor), Université Claude Bernard Lyon 1

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RESEARCH TOPICS

Security of SDN control. Introducing security into the SDN Controller Placement Problem, in particular for O-RAN architectures, and taking into account anomaly detection when placing controllers.

Observation of SDN control. Observing the activity of SDN controllers to detect anomalies in their decisions, from a single controller to distributed controllers.

Privacy of geolocation systems. Analysis of the privacy risks of geolocation ecosystems and design of privacy-preserving geolocation and tracking architectures.

Optimization in wireless networks. Energy saving in Wi-Fi networks through Target Wake Time (TWT), and control of a drone fleet for network coverage using reinforcement learning.

EXPERIENCE

Université Claude Bernard Lyon 1, LIP Sep 2023 – Now
Maître de conférences (Associate Professor). Computer Science department and LIP, HoWNeT team.

Université Côte d'Azur, I3S Oct 2022 – Aug 2023
Attaché Temporaire d'Enseignement et de Recherche (ATER). Teaching at the IUT R&T (Networks and Telecommunications); research in the Signals, Images and Systems (SIS) team, with collaborations with the COMRED and SPARKS teams.

Université de Lorraine, CRAN Oct 2019 – Sep 2022
Doctoral researcher, Eco-Technical Systems Engineering (ISET) team. Doctoral contract funded by the Lorraine Université d'Excellence initiative (LUE I-SITE), IMPACT project “DigiTrust”.

EDF R&D, Chatou Feb 2019 – Jul 2019
Research intern, P11 group (systems engineering and control). Dependability of programmed systems.

Technical University of Munich Sep 2017 – Feb 2018
Research intern, Embedded Systems Safety. Supervisory control theory (SCT).

EDUCATION

Université de Lorraine, CRAN 2019 – 2022
Ph.D. in Computer Engineering (CNU sections 27/61), ISET team. Defended publicly on 17 November 2022.
Thesis title: “Détection d'anomalies de sûreté et sécurité d'un contrôle centralisé de réseau” (in French).
Advisors: Prof. T. Divoux and Prof. J.-P. Georges (Université de Lorraine).
Funding: doctoral contract of the Lorraine Université d'Excellence initiative (LUE I-SITE), IMPACT project “DigiTrust”.
Committee: D. Maquin (Prof., Université de Lorraine), president; P. Berthou (MCF HDR, Université Toulouse III), I. Demongodin (Prof., Aix-Marseille Université) and D. Magoni (Prof., Université de Bordeaux), reviewers; Y.-Q. Song (Prof., Université de Lorraine) and S. Souihi (MCF, Université Paris-Est Créteil), examiners; T. Divoux and J.-P. Georges (Prof., Université de Lorraine), examiners and advisors; J.-J. Lesage (Prof., ENS Paris-Saclay), guest.

Université Paris-Saclay – ENS Paris-Saclay, LURPA 2018 – 2019
Master 2 Ingénierie des Systèmes Complexes (ISC), research track “Commande et Conception de Systèmes Critiques”, supervised by Prof. J.-J. Lesage.
Master thesis: “Formalism and semantics of PyCATSHOO: A simulator of distributed stochastic hybrid automata” (advisors P.-Y. Piriou and T. Lemattre, EDF R&D), published in Reliability Engineering & System Safety (JCR, Q1).

Institut Supérieur de Mécanique de Paris (Supméca) 2016 – 2019
Engineering school (Paris): full training in mechanics, then specialization in logistics and mechatronics.

SELECTED PUBLICATIONS

Detection-Aware Controller Placement in Software-Defined Networks
L. Desgeorges, F. Bronzino, and F. Faticanti.
In IEEE IWCMC 2026, pp. 138–143.

SoK: Mapping the Privacy Landscape of Geolocation Ecosystems
A. Laouar, P. Lachat, L. Desgeorges, M. Cunche, V. Roca, P. Vicat-Blanc, and F. Bronzino.
Proceedings on Privacy Enhancing Technologies (PoPETs) 2026(4), pp. 455–479. Presented at PETS 2026, Calgary, Canada.

PMSA: Power Mode Selection Algorithm in IEEE 802.11 WLANs operating TWT
L. Desgeorges, T. Begin, and I. Guérin Lassous.
In ACM/IEEE MSWiM 2025, Barcelona, Spain, pp. 252–261. *Selected among the three best papers of the conference.*

Detection of anomalies of a non-deterministic software-defined networking control
L. Desgeorges, J.-P. Georges, and T. Divoux.
Computers & Security 129 (2023).

RSC to the ReSCu: Automated Verification of Systems of Communicating Automata
L. Desgeorges and L. Germerie Guizouarn.
In COORDINATION 2023, pp. 135–143. Springer.

A technique to monitor threats in SDN data plane computation
L. Desgeorges, J.-P. Georges, and T. Divoux.
In IEEE HPSR 2021, pp. 1–6.

Formalism and semantics of PyCATSHOO: A simulator of distributed stochastic hybrid automata
L. Desgeorges, P.-Y. Piriou, T. Lemattre, and H. Chraibi.
Reliability Engineering & System Safety 208 (2021).

PROJECTS (AS A MEMBER)

ANR GTTP: Global Tracking Transfer Protocol 2025 – 2028
Project developing a unified, privacy-first geolocation infrastructure. With Inria and Sentiens.

ANR WISENET: Wi-Fi Energy-Saving Solutions for Uncoordinated Networks AAPG 2026
Energy-saving solutions for clusters of adjacent, possibly uncoordinated Wi-Fi networks, to reduce their global energy consumption while maintaining throughput and latency.

TEACHING

Université Claude Bernard Lyon 1, Computer Science Department 2023 – Now
L1: LIFRW.
L3: LIFRES; RS2P (Networks, Systems and Security through Practice).
M2 SRS: BEPC (Bibliography, Case Study, Project, Certification); RA (SDN, VNF, Network Slicing, BGP); ASR (Resources, Planning).

Université Côte d'Azur, IUT R&T 2022 – 2023
Networks and Telecommunications.

Université de Lorraine 2019 – 2022
Teaching assignments during the Ph.D.

ADVISING

Ph.D. students

Soumaya Alaoui 2026 – Now
“Distributed Observation of Distributed SDN Controllers for Anomaly Detection”. With F. Bronzino and I. Guérin Lassous.

Augustin Laouar 2025 – Now
“Privacy-aware geolocation”. With F. Bronzino.

Research interns

Maxime Gautrot (M2, 2026), Dongyu Zhang (M2, 2026), Alexis Carré (M2, 2026), Kevin Andrianasolo Lala (M2, 2026), Dorian Tonnis (M1, 2025), Maya Al Farra (M2, 2025), Andrei Danila (Erasmus Mundus thesis, 2024), Kuider Benzait (L3, 2024), Mirwis Satarzai (L3, 2024), Ahmed Marzouki (L2, 2024).

AWARDS AND HONORS

- Paper selected among the three best papers of ACM/IEEE MSWiM 2025.
- Doctoral contract of the Lorraine Université d'Excellence initiative (LUE I-SITE), IMPACT project “DigiTrust”, 2019.

PROFESSIONAL SERVICE

Technical Program Committee Member: IEEE ICCCN 2027; IEEE IWCMC (Sensors and IoT track).

Shadow Program Committee Chair: AlgoTel/CoRes.

Organization Committee Member: 6th IFAC Symposium on Telematics Applications (TA22), Nancy, June 2022.

ACADEMIC SERVICE

Elected member of the Computer Science Department council, UCBL1 2026 – Now

Elected member of the CRAN laboratory council Sep 2020 – Sep 2022

Representative of about 150 Ph.D. students and postdocs. Organized the Ph.D. students' welcome days (scientific morning and team-building afternoon, 3,000 € budget) and represented them before the HCERES 2022 evaluation committee, after running a survey among the laboratory's doctoral students.

Elected member of the CRAN council board (Bureau du conseil) Sep 2021 – Sep 2022

Ph.D. students' representative, preparing the agenda of the laboratory council.

Hiring committee member:

Maître de conférences in computer science at Université de Lorraine 2025

Maître de conférences in computer science at Université Côte d'Azur. 2026

Ph.D. follow-up committee (CSI) member:

- Thomas Le Tarnec, Université de Strasbourg (supervised by J. Montavont and T. Noël), with G. Le Gall (Université de Rennes) and G. Z. Papadopoulos (IMT Atlantique).
- Lu Yi, Université Côte d'Azur (supervised by G. Urvoy-Keller and R. Aparicio-Pardo).